XISOSIS · Institutional Research Paper

J-INVARIANT SUBGROUPS OF SU(2^n): REPLACING LIE-ALGEBRA DIMENSION

Main result: Problem 3 is answered in the negative. Efficient classical sampling of $\mathbb{Q}$-structured circuits is not equivalent to polynomial Lie-algebra dimension: an explicit family with Lie algebra of dimension $3 \cdot 2^{n-2}$ is sampled exactly in linear time, and padded IQP circuits with abelian Lie algebra cannot be sampled efficiently unless the polynomial hierarchy collapses. Solutions of Exercises 3, 6, 7 and 10 are included.

Roger Cooperman

https://orcid.org/0009-0009-5011-2135

and A.E. Petrunina V.V.

https://orcid.org/0009-0002-4595-736X

DOI: https://doi.org/10.5281/zenodo.23769211

GitHub repository:

https://github.com/teknoquants/J-INVARIANT-SUBGROUPS-OF-SU-2n-REPLACING-LIE-ALGEBRA-DIMENSION

Vienna · 29 October 2026

© 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License (see Licence)

Abstract

This paper studies circuits, operator algebras and lattice Hamiltonians that respect the canonical antiunitary $J_n = (\sigma_y)^{\otimes n} K$, whose square $J_n^2 = (-1)^n I$ alternates between a quaternionic and a real sector with the parity of n. We prove the structural facts that govern this $\mathbb{Q}$-structure:

- the odd-weight criterion for $\mathbb{Q}$-structured rotations, and its even-weight counterpart for $\mathbb{Q}$-symmetric Hamiltonians;
- Kramers degeneracy for odd n;
- the dimension formula $\dim \mathfrak{k}_n = (4^n - (-2)^n)/2$;
- a degree rule for Majorana monomials;
- a block factorisation theorem and a code criterion for classical sampling.

Answer to Problem 3. We answer Problem 3 of [1] in the negative: for $\mathbb{Q}$-structured circuits, efficient classical sampling is *not* equivalent to polynomial dimension of the dynamical Lie algebra.

- *Necessity fails, unconditionally.* An explicit family of $n-1$ odd-weight generators has a dynamical Lie algebra of dimension $3 \cdot 2^{n-2}$, yet every circuit from it can be sampled exactly in time linear in its length (Theorem 3.7).
- *Sufficiency fails unless the polynomial hierarchy collapses.* Padding the commuting IQP circuits of Bremner, Jozsa and Shepherd with one ancilla gives $\mathbb{Q}$-structured circuits with an abelian dynamical Lie algebra of dimension $n(n+1)/2$, whose efficient classical sampling would collapse the polynomial hierarchy to its third level (Theorem 3.9).

A parity-padding lemma shows that $\mathbb{Q}$-structure costs one qubit and nothing else (Lemma 3.8). The Lie algebra alone therefore cannot decide sampling complexity (Corollary 3.10). We state a refined, one-directional question in its place (Question 3.13).

We give complete solutions of Exercises 3, 6, 7 and 10 of the Cooperman–Petrunina Challenge Series [1], each quoted verbatim before its solution. In particular, for Exercise 7: the commuting weight-3 circuits are $\mathbb{Q}$-structured, have an abelian dynamical Lie algebra of dimension $n-2$, are trivially sampleable, and are not quadratic in fermions. We recover the complexification identity $L_{\mathbb{H}}(F_5) \otimes_{\mathbb{R}} \mathbb{C} \cong L_{\mathbb{R}}(F_2) \otimes_{\mathbb{R}} \mathbb{C}$. Finally, we prove that a J-compatible isometric embedding of qubit registers exists exactly when the number of qubits grows by an even number.

Scope. Problem 3 is answered as above; its sufficiency half is conditional on the standard assumption that the polynomial hierarchy is infinite. The central questions of Problems 1 and 2 remain open: β-invariance of free entropy dimension, reflection positivity under $\mathbb{Q}$-structure, and the parity independence of the continuum limit. They are stated as conjectures or questions (Conjectures 4.5, 5.5 and 5.7, Question 5.3). Appendix A lists every result of the earlier draft that this version corrects or withdraws.

Problem and exercises addressed

The wording below is quoted verbatim from Sections 4 and 5 of [1], with the mathematics set in the same notation. Each item is quoted again, in full, where it is solved. Reference numbers inside quotations refer to the bibliography of [1].

Item	Where it is solved	Status in this paper
Problem 3 · The $\mathbb{Q}$-Structured Simulability Dichotomy	Section 3.2, Theorem 3.11	Answered: **no**. Necessity fails unconditionally (Theorem 3.7); sufficiency fails unless the polynomial hierarchy collapses (Theorem 3.9)
Exercise 3 ($\star$)	Section 2, Lemma 2.1	Solved

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

Item	Where it is solved	Status in this paper
Exercise 7 (⋆⋆)	Section 2, Solution of Exercise 7	Solved
Exercise 6 (⋆⋆)	Section 4, Proposition 4.3	Solved, given the Dykema–Rădulescu formula
Exercise 10 (⋆⋆⋆)	Section 1, Corollary 1.6, and Section 5, Proposition 5.4	Solved

1. The canonical antiunitary framework

Definition 1.1 (canonical antiunitary operator). Let K be complex conjugation in the computational basis of $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, and $S_n = (\sigma_y)^{\otimes n}$. The canonical antiunitary operator is

$$J_n := S_n K.$$

Theorem 1.2 (parity of J_n^2). J_n is antiunitary and

$$J_n^2 = (-1)^n I.$$

So J_n is an involution for even n; for odd n it satisfies $J_n^2 = -I$ (quaternionic type).

Proof. S_n is unitary and K is antiunitary, so J_n is antiunitary. Since $KAK = \overline{A}$,

$$J_n^2 = S_n K S_n K = S_n \overline{S_n} = (\sigma_y)^{\otimes n}(-\sigma_y)^{\otimes n} = (-1)^n \left(\sigma_y^2\right)^{\otimes n} = (-1)^n I. \qquad \blacksquare$$

Definition 1.3 ($\mathbb{Q}$-structured unitaries). The $\mathbb{Q}$-structured group is $K_n = \{U \in U(2^n) : J_n U J_n^{-1} = U\}$, and its Lie algebra is $\mathfrak{k}_n = \{A \in \mathfrak{su}(2^n) : J_n A J_n^{-1} = A\}$.

Lemma 1.4 (odd-weight criterion). Let $P \neq I$ be a Pauli string of weight $w(P)$. Then

$$J_n P J_n^{-1} = (-1)^{w(P)} P,$$

and $\exp(-i\theta P/2) \in K_n$ **for all** $\theta \in \mathbb{R}$ if and only if $w(P)$ is odd.

Proof. For one qubit, $\sigma_y \overline{\sigma_\alpha} \sigma_y = -\sigma_\alpha$ for $\alpha \in \{x, y, z\}$ and $\sigma_y I \sigma_y = I$; taking tensor products gives the first claim. Since J_n is antilinear,

$$J_n \exp(-i\theta P/2) J_n^{-1} = \exp(\tfrac{i\theta}{2}\, J_n P J_n^{-1}) = \exp(\tfrac{i\theta}{2}(-1)^{w(P)} P).$$

This equals $\exp(-i\theta P/2)$ for all θ if and only if the derivatives at $\theta = 0$ agree, that is $(-1)^{w(P)} = -1$. $\blacksquare$

The qualifier "for all θ" is needed: at $\theta = 2\pi$ every rotation equals $-I$, which lies in K_n whatever the weight.

Corollary 1.5 ($\mathbb{Q}$-symmetric Hamiltonians). A Hermitian $H = \sum_P c_P P$ (with $c_P \in \mathbb{R}$) satisfies $[H, J_n] = 0$ if and only if $c_P = 0$ for every odd-weight P. Consequently $\mathfrak{k}_n = \operatorname{span}_{\mathbb{R}}\{iP : w(P) \text{ odd}\}$.

Proof. The coefficients are real, so antilinearity leaves them unchanged and $J_n H J_n^{-1} = \sum_P c_P (-1)^{w(P)} P$. For $A = \sum_P a_P\, iP$ with $a_P \in \mathbb{R}$, antilinearity gives $J_n A J_n^{-1} = \sum_P a_P (-i)(-1)^{w(P)} P$, which equals A exactly when every term has odd weight. $\blacksquare$

Generators of $\mathbb{Q}$-structured rotations therefore have odd weight, while $\mathbb{Q}$-symmetric Hamiltonians have even weight. Section 5 uses the second fact.

> **Exercise 10 (⋆⋆⋆)** [1, Section 5]. Explain why, for odd n, a J_n-symmetric Hamiltonian has every energy level at least doubly degenerate, and why Problem 1 therefore measures the mass gap from the ground multiplet rather than the ground state.

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

Corollary 1.6 (Kramers degeneracy). If n is odd and $[H, J_n] = 0$, every eigenvalue of H has even multiplicity. If n is even, no degeneracy is forced, and none is excluded.

Proof. Let $Hv = Ev$. Then E is real, so $H(J_n v) = J_n H v = E\, J_n v$. Antiunitarity gives $\langle J_n x, J_n y\rangle = \langle y, x\rangle$. With $x = v$ and $y = J_n v$, this yields $\langle J_n v, J_n^2 v\rangle = \langle J_n v, v\rangle$, that is $-\langle J_n v, v\rangle = \langle J_n v, v\rangle$. Hence $J_n v \perp v$, and J_n makes each eigenspace a quaternionic vector space, of even complex dimension. For even n: a generic even-weight H on 4 qubits has a simple spectrum, while $H = -Z_1 Z_2$ on 2 qubits is J_2-symmetric with a doubly degenerate ground level (Appendix B). ■

Solution of Exercise 10. *First part.* By Theorem 1.2, $J_n^2 = -I$ for odd n. Corollary 1.6 then shows that every energy level of a J_n-symmetric Hamiltonian has even multiplicity, so it is at least doubly degenerate. *Second part.* In particular the ground level is degenerate: $E_0 = E_1$ for every odd-n $\mathbb{Q}$-symmetric Hamiltonian. The difference $E_1 - E_0$ is therefore always 0 and cannot serve as a mass gap. The meaningful gap is the distance from the whole ground multiplet to the next distinct level, $\Delta = E_{\text{next}} - E_0$. This is why Problem 1 measures the mass gap from the ground multiplet rather than from the ground state (Proposition 5.4(2)).

2. Dimension and fermionic degree

> **Exercise 3 (⋆)** [1, Section 5]. Show that the number of odd-weight Pauli strings on n qubits is $(4^n - (-2)^n)/2$, and check that this equals $\dim \mathfrak{sp}(2^{n-1})$ for odd n and $\dim \mathfrak{so}(2^n)$ for even n.

Lemma 2.1 (dimension of $\mathfrak{k}_n$; solution of Exercise 3).

$$\dim \mathfrak{k}_n = \sum_{w \text{ odd}} \binom{n}{w} 3^w = \frac{4^n - (-2)^n}{2},$$

and $\mathfrak{k}_n \cong \mathfrak{sp}(2^{n-1})$ for odd n, $\mathfrak{k}_n \cong \mathfrak{so}(2^n)$ for even n.

Proof. By Corollary 1.5, $\mathfrak{k}_n$ has a basis of odd-weight strings; there are $\binom{n}{w}3^w$ of weight w. Subtracting $(1-3)^n = \sum_w \binom{n}{w}(-3)^w$ from $(1+3)^n$ leaves twice the odd terms. The unitaries commuting with an antiunitary J on $\mathbb{C}^N$ form $Sp(N/2)$ if $J^2 = -I$ and a conjugate of $O(N)$ if $J^2 = +I$ [2]. Theorem 1.2 decides which case applies, and the dimensions agree:

$$\frac{4^n + 2^n}{2} = 2^{n-1}(2^n{+}1) = \dim \mathfrak{sp}(2^{n-1}) \ (n \text{ odd}), \qquad \frac{4^n - 2^n}{2} = 2^{n-1}(2^n{-}1) = \dim \mathfrak{so}(2^n) \ (n \text{ even}).$$ ■

Solution of Exercise 3. The first part of the exercise (the count) is the displayed identity of Lemma 2.1, proved by the binomial argument. The second part (the check against $\mathfrak{sp}$ and $\mathfrak{so}$) is the last display of the proof. For small n:

n	Odd-weight strings $\sum_{w \text{ odd}} \binom{n}{w}3^w$	$(4^n - (-2)^n)/2$	Lie algebra	Its dimension
1	3	3	$\mathfrak{sp}(1)$	$1 \cdot 3 = 3$
2	6	6	$\mathfrak{so}(4)$	$4 \cdot 3/2 = 6$
3	36	36	$\mathfrak{sp}(4)$	$4 \cdot 9 = 36$
4	120	120	$\mathfrak{so}(16)$	$16 \cdot 15/2 = 120$
5	528	528	$\mathfrak{sp}(16)$	$16 \cdot 33 = 528$

Convention (Jordan–Wigner, Z on site). Throughout,

$$\gamma_{2j-1} = \Big(\prod_{k<j} Z_k\Big) X_j, \qquad \gamma_{2j} = \Big(\prod_{k<j} Z_k\Big) Y_j, \qquad \{\gamma_r, \gamma_s\} = 2\delta_{rs} I.$$

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

For a set $A = \{a_1 < \cdots < a_p\}$ of indices write $\gamma_A = \gamma_{a_1} \cdots \gamma_{a_p}$, a monomial of degree $|A| = p$.

Lemma 2.2 (Majorana degrees).

1. $Z_j = -i\gamma_{2j-1}\gamma_{2j}$, of degree 2.
2. $Z_jZ_{j+1}Z_{j+2} = i\,\gamma_{2j-1}\gamma_{2j}\gamma_{2j+1}\gamma_{2j+2}\gamma_{2j+3}\gamma_{2j+4}$, of degree 6.
3. $X_j = (\prod_{k<j} Z_k)\gamma_{2j-1}$ has degree $2j-1$; single-site X_j is not quadratic in this convention.
4. In the alternative convention with X on site, where X_j is quadratic, $Z_2Z_3Z_4 = -\gamma_1\gamma_2\gamma_3\gamma_6\gamma_7$ has degree 5. So $Z_jZ_{j+1}Z_{j+2}$ is quadratic in neither convention.

Proof. (1) The strings cancel: $\gamma_{2j-1}\gamma_{2j} = X_jY_j = iZ_j$. (2) Multiply three copies of (1). The index sets are disjoint, so no factor cancels, and $(-i)^3 = i$. (3) Each Z_k with $k < j$ is a product of two Majoranas by (1). (4) This is a direct computation, recorded in Appendix B. ■

> **Exercise 7 (⋆⋆)** [1, Section 5]. Show that circuits generated by the commuting weight-3 strings $Z_jZ_{j+1}Z_{j+2}$ are $\mathbb{Q}$-structured, have an abelian dynamical Lie algebra of dimension $n-2$, and are trivially sampleable from $|0\,..\,0\rangle$. Show that these generators are not quadratic in fermions under the Jordan–Wigner transformation. Explain why this rules out "free fermions are the only tractable case" as a conjecture.

Proposition 2.3 (solution of Exercise 7). Let $n \geq 3$ and $\mathcal{G}_{ZZZ} = \{Z_jZ_{j+1}Z_{j+2}\}_{j=1}^{n-2}$. The five parts of the exercise are answered in order.

1. $\mathbb{Q}$*-structured.* Each $Z_jZ_{j+1}Z_{j+2}$ has weight 3, which is odd, so by Lemma 1.4 every rotation $\exp(-i\theta Z_jZ_{j+1}Z_{j+2}/2)$ lies in K_n for all θ. Every circuit built from them is therefore $\mathbb{Q}$-structured.
2. *Abelian DLA of dimension* $n-2$. All generators are diagonal in the computational basis, so they commute and every bracket vanishes. The DLA is the real span of the $n-2$ operators $iZ_jZ_{j+1}Z_{j+2}$. These are distinct Pauli strings, hence linearly independent, so
$$\dim \mathfrak{g}_{ZZZ} = n-2.$$
3. *Trivially sampleable from* $|0\cdots0\rangle$. Since $Z_k|0\rangle = |0\rangle$, each generator fixes $|0\cdots0\rangle$, and
$$\prod_\ell \exp(-i\theta_\ell Z_{j_\ell}Z_{j_\ell+1}Z_{j_\ell+2}/2)\,|0\cdots0\rangle = e^{-i\sum_\ell \theta_\ell/2}\,|0\cdots0\rangle.$$
The output is $0\cdots0$ with probability one, for all angles, so a sampler just outputs that string.
4. *Not quadratic in fermions.* Under the Jordan–Wigner transformation each Z_k is the quadratic $-i\gamma_{2k-1}\gamma_{2k}$ (Lemma 2.2(1)). The product over three distinct sites is the degree-6 monomial
$$Z_jZ_{j+1}Z_{j+2} = i\,\gamma_{2j-1}\gamma_{2j}\gamma_{2j+1}\gamma_{2j+2}\gamma_{2j+3}\gamma_{2j+4}$$
(Lemma 2.2(2)). A monomial on six distinct Majoranas is not a linear combination of quadratic monomials, since distinct monomials are linearly independent. This holds for every ordering of the qubits, because each Z_k stays on-site. In the alternative convention with X on site the generator has degree 5 (Lemma 2.2(4)), again not quadratic.
5. *Why this rules out the conjecture.* The conjecture "free fermions are the only tractable case" asserts that every efficiently sampleable family of $\mathbb{Q}$-structured circuits is a free-fermion family, with generators quadratic in Majoranas. Parts 1–4 exhibit a $\mathbb{Q}$-structured family that is efficiently sampleable (part 3) but whose generators are not quadratic (part 4). This single family is a counterexample, so the conjecture is false. Tractability within the $\mathbb{Q}$-structured class is therefore not captured by fermionic degree. Section 3.2 shows that Lie-algebraic dimension is not the dividing line either (Theorem 3.11). ■

3. The simulability dichotomy (Problem 3)

> **Problem 3 · The $\mathbb{Q}$-Structured Simulability Dichotomy** [1, Section 4]. *Full statement, quoted verbatim; reference numbers and section numbers inside the quota-*

tion refer to [1].

Audience. Structural complexity theorists, quantum algorithm designers, Lie theorists and algebraic geometers.

The question. For circuits built from odd-weight Pauli rotations, starting in $|0 \dots 0\rangle$ and measured in the computational basis, is efficient classical sampling (for all angles) equivalent to the dynamical Lie algebra having dimension polynomial in n?

Why it matters. It asks whether Lie-algebra dimension is the exact dividing line between classically simulable and genuinely quantum computation inside a natural symmetry class. It also sharpens the question raised by Cerezo et al. [6] of whether provable absence of barren plateaus implies classical simulability.

What is already known. Free-fermion (matchgate) circuits are classically simulable [26, 27, 28]. The g-sim method simulates expectation values whenever the Lie algebra has polynomial dimension [7], but sampling the full output distribution is a different task. Sampling hardness for commuting circuits rests on non-collapse of the polynomial hierarchy [29]. Dynamical Lie algebras of 2-local spin systems are classified in [30].

Entry points

- Construct a family with superpolynomial Lie algebra that is nevertheless efficiently sampleable. This would disprove the conjecture and needs no complexity breakthrough.
- Show that polynomial dimension implies efficient sampling, beyond the expectation values that g-sim provides.
- Classify the polynomial-dimension Lie algebras that sets of odd-weight Pauli strings can generate, and the gradient-variance scaling of each.

Prerequisites

Quantum circuits, the Jordan–Wigner transformation, basic Lie algebras, and complexity classes up to PSPACE and the polynomial hierarchy.

Reading path

- Start: McClean et al. [3]; Ragone et al. [4]; Section 3.6 above.
- Core: Valiant [26]; Jozsa–Miyake [28]; Goh et al. [7]; Bremner–Jozsa–Shepherd [29].
- Frontier: Cerezo et al. [6]; Wiersema et al. [30]; white paper 3 of [1].

What this section establishes. Section 3.1 sets up the problem and proves supporting results. Section 3.2 answers it: the equivalence fails in both directions. The failure of necessity is unconditional and follows entry point 1 above. The failure of sufficiency holds unless the polynomial hierarchy collapses, which is the standard assumption under which sampling problems are called classically hard.

3.1 Set-up and supporting results

Definition 3.1. For a set $\mathcal{G}$ of Hermitian generators, $\mathfrak{g} = \langle iH : H \in \mathcal{G} \rangle_{\text{Lie}}$ is the dynamical Lie algebra (DLA). $\mathcal{C}_{\text{sample}}(\mathcal{G})$ denotes the classical time needed to sample, to total variation distance ε, from the computational-basis output of any circuit $\prod_\ell \exp(-i\theta_\ell H_\ell)\,|0 \cdots 0\rangle$ with $H_\ell \in \mathcal{G}$.

Theorem 3.2 (known results).

1. If every generator is a quadratic Majorana monomial, then $\dim \mathfrak{g} = O(n^2)$ and the output can be sampled in $\text{poly}(n)$ time (matchgate circuits [3, 4, 5]).
2. If $\dim \mathfrak{g} = \text{poly}(n)$, expectation values $\langle O \rangle$ of observables $O \in i\mathfrak{g}$ can be computed in time

polynomial in $\dim \mathfrak{g}$, under the conditions of [6] (g-sim).

Neither statement gives sampling from polynomial DLA dimension in general: g-sim computes expectation values, not output distributions.

Statement 3.3 (the dichotomy as posed in Problem 3 of [1]). Let $\mathcal{G}$ consist of odd-weight Pauli strings.

- (a) *Sufficiency.* If $\dim \mathfrak{g} = \mathrm{poly}(n)$, then $\mathcal{C}_{\mathrm{sample}}(\mathcal{G}) = \mathrm{poly}(n)$.
- (b) *Necessity.* If $\mathcal{C}_{\mathrm{sample}}(\mathcal{G}) = \mathrm{poly}(n)$ for all angles, then $\dim \mathfrak{g} = \mathrm{poly}(n)$.

Problem 3 asks whether (a) and (b) both hold. Theorem 3.11 shows that (b) is false and that (a) is false unless the polynomial hierarchy collapses. The earlier draft's "Theorem 3.2" asserted (a) and claimed to refute (b), but its argument was invalid (Appendix A); the correct counterexample is Theorem 3.7.

Lemma 3.4 (degree rule). For index sets A, B with $|A| = p$, $|B| = q$ and $s = |A \cap B|$,

$$\gamma_A \gamma_B = (-1)^{pq-s}\, \gamma_B \gamma_A, \qquad \gamma_A \gamma_B = \pm\, \gamma_{A \triangle B}.$$

Hence $[\gamma_A, \gamma_B] = 0$ when $pq - s$ is even. Otherwise $[\gamma_A, \gamma_B] = \pm 2\, \gamma_{A \triangle B}$, of degree $p + q - 2s$.

Proof. Move each γ_b, $b \in B$, from the right of γ_A to its left. It anticommutes with the $p - [b \in A]$ factors of γ_A different from itself, and commutes with itself. The total sign is $(-1)^{pq-s}$. In the product, each shared index appears twice and squares to I after reordering, which leaves $\pm\gamma_{A \triangle B}$. ■

Corollary. Quadratic monomials ($p = q = 2$) close under brackets: the bracket has degree $2 + 2 - 2 = 2$ when $s = 1$ and vanishes when $s \in \{0, 2\}$. Degree is not preserved beyond the quadratic case. Two degree-6 monomials sharing one index have a nonzero bracket of degree $6 + 6 - 2 = 10$. So the span of degree-6 monomials is not closed under brackets.

Proposition 3.5 (probabilities need all degrees). For $x \in \{0, 1\}^n$,

$$|x\rangle\langle x| = 2^{-n} \sum_{S \subseteq \{1, \ldots, n\}} (-1)^{x \cdot S}\, Z_S, \qquad Z_S = \prod_{j \in S} Z_j = (-i)^{|S|} \prod_{j \in S} \gamma_{2j-1} \gamma_{2j},$$

so $P(x) = 2^{-n} \sum_S (-1)^{x \cdot S} \langle Z_S \rangle$. The monomials have degree $2|S|$, and every even degree $0, 2, \ldots, 2n$ occurs, among 2^n terms.

Proof. Expand $\prod_j \frac{1}{2}(I + (-1)^{x_j} Z_j)$ and apply Lemma 2.2(1). ■

A method that computes $\langle O \rangle$ only for O in a polynomial-dimensional space therefore does not, by itself, give $P(x)$. Theorem 3.9 shows that, in general, it cannot: there are circuits with abelian DLA for which every expectation value $\langle iP \rangle$, $iP \in \mathfrak{g}$, is trivial to compute, yet sampling is hard unless the polynomial hierarchy collapses.

Theorem 3.6 (block factorisation). Suppose $\mathcal{G} = \mathcal{G}_1 \sqcup \cdots \sqcup \mathcal{G}_m$, where the qubit supports $S_k = \bigcup_{H \in \mathcal{G}_k} \mathrm{supp}(H)$ are pairwise disjoint. Then, for every circuit over $\mathcal{G}$:

1. $U = \bigotimes_k U_k$ with U_k acting on S_k, and $P(x) = \prod_k P_k(x_{S_k}) \cdot \prod_{j \notin \cup S_k} \delta_{x_j, 0}$;
2. $\dim \mathfrak{g} = \sum_k \dim \mathfrak{g}_k \le \sum_k 4^{|S_k|}$;
3. sampling costs $\sum_k O(L\, 4^{|S_k|})$ for a circuit of L gates, by simulating each block separately.

If $|S_k| \le c \log n$ for every k, both the sampling cost and $\dim \mathfrak{g}$ are polynomial in n.

Proof. Gates from different blocks act on disjoint qubits and commute. Collecting the gates of each block gives $U = \bigotimes_k U_k$, which leaves untouched qubits in $|0\rangle$ and factorises the distribution. The DLA is the direct sum of the block DLAs, and $\mathfrak{g}_k \subseteq \mathfrak{u}(2^{|S_k|})$. Each block state vector has $2^{|S_k|}$ entries, and each gate updates it in $O(4^{|S_k|})$ time. ■

Remarks.

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

- A disconnected commutation graph is not enough. X_1X_2 and Z_1Z_2 commute but share support, and from $|00\rangle$ the output $(\cos^2\frac{\theta}{2}, 0, 0, \sin^2\frac{\theta}{2})$ does not factorise over the qubits (Appendix B).
- Theorem 3.6 always yields polynomial DLA dimension. Theorem 3.12 below gives a sufficient condition that does not.

3.2 Answer to Problem 3

Theorem 3.7 (efficient sampling with exponential Lie algebra). For $n \geq 3$, let qubits $1, \ldots, n-1$ be *controls* and qubit n the *target*, and let

$$\mathcal{G}_n^A = \{X_n,\ Y_n\} \cup \{Z_iZ_{i+1}X_n : 1 \leq i \leq n-2\}.$$

1. Every generator has odd weight (1 or 3), so every circuit over $\mathcal{G}_n^A$ is $\mathbb{Q}$-structured.
2. The dynamical Lie algebra is

$$\mathfrak{g}_n^A = \mathrm{span}_{\mathbb{R}}\{\, iZ^s \otimes \sigma_n : s \in E,\ \sigma \in \{X, Y, Z\}\,\} \cong \mathfrak{su}(2)^{\oplus 2^{n-2}}, \qquad \dim \mathfrak{g}_n^A = 3 \cdot 2^{n-2},$$

   where $E \subseteq \mathbb{F}_2^{n-1}$ is the set of even-weight vectors and $Z^s = \prod_i Z_i^{s_i}$ on the controls.
3. For every circuit $U = \prod_{\ell=1}^{L} \exp(-i\theta_\ell P_\ell/2)$ with $P_\ell \in \mathcal{G}_n^A$ and all angles,

$$U|0\cdots 0\rangle = |0\cdots 0\rangle_{\text{controls}} \otimes V|0\rangle_n, \qquad V = \prod_{\ell=1}^{L} \exp(-i\theta_\ell\, \sigma(P_\ell)/2),$$

   where $\sigma(P_\ell) \in \{X, Y\}$ is the target factor of P_ℓ. The output distribution is therefore computed exactly in $O(L)$ arithmetic operations, and sampled exactly.

So $\mathcal{C}_{\text{sample}}(\mathcal{G}_n^A) = O(L)$ for all angles, while $\dim \mathfrak{g}_n^A$ grows exponentially. Statement 3.3(b) is false.

Proof. (1) X_n and Y_n have weight 1; $Z_iZ_{i+1}X_n$ has weight 3.

(2) *Upper bound.* Each generator has the form $Z^s \otimes \sigma_n$ with $s \in E$ and $\sigma \in \{X, Y\}$. The factor Z^s acts on the controls and commutes with every target operator, so

$$[Z^s \otimes \sigma_n,\ Z^{s'} \otimes \tau_n] = Z^{s+s'} \otimes [\sigma, \tau]_n.$$

E is closed under addition, so $\mathfrak{g}_n^A$ lies in the span of the $3|E|$ strings $iZ^s \otimes \sigma_n$.

Lower bound. $[X_n, Y_n] = 2iZ_n$, so all three σ_n occur with $s = 0$. If $iZ^s \otimes \sigma_n \in \mathfrak{g}_n^A$ for all σ, then bracketing iZ^sZ_n with the generator $iZ_iZ_{i+1}X_n$ gives a multiple of $iZ^{s+e_i+e_{i+1}}Y_n$. Bracketing that with iX_n and iY_n gives the other two Paulis, with the same control pattern. The vectors $e_i + e_{i+1}$, $1 \leq i \leq n-2$, span E, so every $s \in E$ is reached. Distinct Pauli strings are linearly independent, so the dimension is $3|E| = 3 \cdot 2^{n-2}$.

Structure. The span $\mathcal{A}$ of $\{Z^s : s \in E\}$ is a commutative algebra isomorphic to $\mathbb{R}^{|E|}$, with characters $s \mapsto (-1)^{s\cdot y}$. Hence $\mathfrak{g}_n^A = \mathcal{A} \otimes \mathfrak{su}(2) \cong \mathfrak{su}(2)^{\oplus|E|}$.

(3) Each Z^s fixes $|0\cdots 0\rangle$ on the controls, so $(Z^s \otimes \sigma_n)(|0\cdots 0\rangle \otimes |\phi\rangle) = |0\cdots 0\rangle \otimes \sigma|\phi\rangle$. Expanding each $\exp(-i\theta P/2) = \cos\frac{\theta}{2}I - i\sin\frac{\theta}{2}P$ and inducting over the gates gives the stated product form. The controls are always measured as 0, and the target follows a 2-dimensional evolution. ■

The controls never change, yet they multiply the dimension of the Lie algebra by 2^{n-2}. Dimension counts the action on the whole Hilbert space, while sampling sees only the orbit of $|0\cdots 0\rangle$.

Lemma 3.8 (parity padding). Let $\mathcal{G}$ be any set of Pauli strings on n qubits, and let qubit $a = n+1$ be an ancilla. Define

$$\mathcal{G}' = \{\, P \otimes Z_a^{[w(P)\ \text{even}]} : P \in \mathcal{G}\,\},$$

appending Z_a exactly to the even-weight strings. Then:

1. every element of $\mathcal{G}'$ has odd weight, so circuits over $\mathcal{G}'$ are $\mathbb{Q}$-structured;
2. for every circuit and all angles, the padded circuit on $|0\cdots0\rangle|0\rangle_a$ outputs the original distribution on the first n qubits and 0 on the ancilla;
3. $\dim \mathfrak{g}(\mathcal{G}) \le \dim \mathfrak{g}(\mathcal{G}') \le 2 \dim \mathfrak{g}(\mathcal{G})$.

Proof. (1) Appending Z_a adds 1 to the weight of exactly the even-weight strings. (2) No element of $\mathcal{G}'$ has X or Y on a, and $Z_a|0\rangle = |0\rangle$, so on the subspace with the ancilla in $|0\rangle$ each padded string acts as the original one. (3) Every element of $\mathfrak{g}(\mathcal{G}')$ is block-diagonal with respect to the ancilla. Compression to the block $a = 0$ is a Lie homomorphism, and it maps generators to generators, so it is onto $\mathfrak{g}(\mathcal{G})$; this gives the lower bound. Compression to the block $a = 1$ maps generators to $\pm$ generators, so it is also onto $\mathfrak{g}(\mathcal{G})$. The two compressions together embed $\mathfrak{g}(\mathcal{G}')$ into $\mathfrak{g}(\mathcal{G}) \oplus \mathfrak{g}(\mathcal{G})$. ■

$\mathbb{Q}$-structure therefore costs one qubit and changes the dynamical Lie algebra dimension by at most a factor of 2. Problem 3 is thus equivalent to the same question for arbitrary Pauli generators.

Theorem 3.9 (polynomial Lie algebra, hard sampling). Let

$$\mathcal{G}_n^B = \{X_j : 1 \le j \le n\} \cup \{X_j X_k Z_a : 1 \le j < k \le n\}$$

on $n+1$ qubits, with ancilla $a = n+1$.

1. Every generator has odd weight; all generators commute; $\mathfrak{g}_n^B$ is abelian, of dimension $n(n+1)/2$.
2. Every IQP circuit of Bremner, Jozsa and Shepherd [12] on n qubits (1- and 2-qubit gates diagonal in the X basis) equals, up to a global phase and on the ancilla-0 sector, a circuit over $\mathcal{G}_n^B$. The same holds for the Ising IQP circuits of Bremner, Montanaro and Shepherd [13], which have terms X_j and $X_j X_k$ with angles in $(\pi/8)\mathbb{Z}$.
3. If every circuit over $\mathcal{G}_n^B$, for all angles, could be sampled classically in polynomial time to within multiplicative error $c < \sqrt{2}$, then PostBPP = PP and the polynomial hierarchy collapses to its third level [12].
4. If such circuits could be sampled classically in polynomial time to within ℓ_1 error $1/192$, the polynomial hierarchy collapses to its third level, assuming the average-case hardness Conjecture 2 of [13].

So, unless the polynomial hierarchy collapses, Statement 3.3(a) is false.

Proof. (1) The weights are 1 and 3. X-strings commute with one another, and Z_a commutes with all of them because no generator has X or Y on a. The DLA is spanned by the $n + \binom{n}{2}$ distinct generators.

(2) A 2-qubit gate diagonal in the X basis is $D = H^{\otimes 2}\,\mathrm{diag}(e^{i\varphi_{00}}, e^{i\varphi_{01}}, e^{i\varphi_{10}}, e^{i\varphi_{11}})\,H^{\otimes 2}$. Its Walsh expansion is

$$D = e^{i\alpha} \exp(i(\beta X_j + \gamma X_k + \delta X_j X_k)),$$

with real $\alpha, \beta, \gamma, \delta$ given by signed averages of the φ's; 1-qubit gates are the case $\gamma = \delta = 0$. The circuit is therefore a product of rotations generated by X_j and $X_j X_k$. Lemma 3.8 pads each $X_j X_k$ to $X_j X_k Z_a$ without changing the output distribution on the first n qubits.

(3) and (4) A classical sampler for $\mathcal{G}_n^B$ yields one, with the same error, for the circuits of [12] and [13]: discard the ancilla bit, which is always 0. The conclusions are Theorem 2 and Corollary 1 of [12] and Theorem 1 of [13]. ■

Corollary 3.10 (the Lie algebra alone cannot decide). Let $\overline{\mathcal{G}}_n^B$ be obtained from $\mathcal{G}_n^B$ by replacing every X with Z. Its circuits are diagonal, so they output $0\cdots0$ with probability one. Yet $\mathfrak{g}(\overline{\mathcal{G}}_n^B) = H^{\otimes n}\,\mathfrak{g}_n^B\,H^{\otimes n}$ (with H on the first n qubits): the two Lie algebras are conjugate, and both abelian of dimension $n(n+1)/2$. Unless the polynomial hierarchy collapses, no property of $\mathfrak{g}$ that is invariant under unitary conjugation decides sampling complexity. Any correct criterion must involve the position of $\mathfrak{g}$ relative to the initial state $|0\cdots0\rangle$ and the computational basis.

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License ·

Theorem 3.11 (answer to Problem 3). For circuits built from odd-weight Pauli rotations, starting in $|0\cdots0\rangle$ and measured in the computational basis:

1. efficient classical sampling for all angles does **not** imply polynomial dimension of the dynamical Lie algebra (Theorem 3.7, unconditional);
2. polynomial dimension of the dynamical Lie algebra, even an abelian one, does **not** imply efficient classical sampling, unless the polynomial hierarchy collapses to its third level (Theorem 3.9).

The equivalence asked for in Problem 3 is therefore false.

Theorem 3.12 (a code criterion for efficient sampling). For a Pauli string P, let $x(P) \in \mathbb{F}_2^n$ mark its X and Y positions, and let $C = \mathrm{span}_{\mathbb{F}_2}\{x(P) : P \in \mathcal{G}\}$. Then for every circuit over $\mathcal{G}$ and all angles, $U|0\cdots0\rangle \in \mathrm{span}\{|c\rangle : c \in C\}$, and the output can be computed exactly in $O(L\, n\, 2^{\dim C})$ time. If $\dim C = O(\log n)$, sampling is efficient, whatever the dimension of $\mathfrak{g}$.

Proof. $P|y\rangle = \lambda\,|y \oplus x(P)\rangle$ with $|\lambda| = 1$ computable in $O(n)$, so each rotation maps $\mathrm{span}\{|c\rangle : c \in y + C\}$ to itself. Start from $y = 0$ and store the $2^{\dim C}$ amplitudes. Each gate updates them in $O(n\, 2^{\dim C})$ time, and the final probabilities are read off directly. ■

Theorem 3.7 is the case $C = \mathrm{span}\{e_n\}$ with $\dim C = 1$. Theorem 3.9 has $\dim C = n$.

Question 3.13 (a refined, one-directional criterion). Let $\mathcal{M}_0$ be the smallest subspace of operators that contains $\rho_0 = |0\cdots0\rangle\langle0\cdots0|$ and is invariant under ad_{iP} for every $P \in \mathcal{G}$. The output state $U\rho_0U^\dagger$ always lies in $\mathcal{M}_0$. Does $\dim \mathcal{M}_0 = \mathrm{poly}(n)$ imply efficient classical sampling?

The converse fails: circuits of single-qubit X rotations produce product states and are trivially sampleable, yet $\dim \mathcal{M}_0 = 3^n$. All examples in this paper are consistent with the question:

Family	$\dim \mathfrak{g}$	$\dim \mathcal{M}_0$ ($n = 3, 4, 5$)	Sampling
Theorem 3.7	$3 \cdot 2^{n-2}$	4, 4, 4	Easy (exact, $O(L)$)
Exercise 7 ($Z_jZ_{j+1}Z_{j+2}$)	$n-2$	1, 1, 1	Easy (deterministic)
Odd-weight X-program (weights 1 and 3)	$n + \binom{n}{3}$	33, 129, 513	IQP-type; not known to be easy
Single-qubit X rotations	n	27, 81, 243 ($= 3^n$)	Easy (product states)

We do not have a proof in either direction; this is the natural successor to Problem 3.

4. Real and quaternionic free group factors (Problem 2)

Definition 4.1. For a real von Neumann algebra $\mathcal{A}$, $\mathcal{F}_{\mathbb{C}}(\mathcal{A}) := \mathcal{A} \otimes_{\mathbb{R}} \mathbb{C}$. Here $L_{\mathbb{R}}(F_k)$ is the real group von Neumann algebra, so that $L_{\mathbb{R}}(F_k) \otimes_{\mathbb{R}} \mathbb{C} \cong L(F_k)$, and $L_{\mathbb{H}}(F_k) := L_{\mathbb{R}}(F_k) \otimes_{\mathbb{R}} \mathbb{H}$. The challenge statement [1] is authoritative for these definitions.

Theorem 4.2 (amplification formula; Dykema [7], Rădulescu [8]). For $1 < r \le \infty$ and $t > 0$,

$$L(F_r)_t \cong L\left(F_{1+\frac{r-1}{t^2}}\right).$$

> **Exercise 6 (⋆⋆)** [1, Section 5]. Using the amplification formula $L(F_r)_t \cong L(F_{1+(r-1)/t^2})$, show that $M_2(L(F_k)) \cong L(F_{(k+3)/4})$. Deduce that $L_{\mathbb{H}}(F_5)$ and $L_{\mathbb{R}}(F_2)$ have the same complexification.

Proposition 4.3 (complexification; solution of Exercise 6). $M_2(L(F_k)) \cong L(F_{(k+3)/4})$, and

$$\mathcal{F}_{\mathbb{C}}(L_{\mathbb{H}}(F_5)) \cong L(F_2) \cong \mathcal{F}_{\mathbb{C}}(L_{\mathbb{R}}(F_2)).$$

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

Proof. $M_2(L(F_k)) \cong L(F_k)_2 \cong L(F_{1+(k-1)/4}) = L(F_{(k+3)/4})$ by Theorem 4.2. Using $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$,

$$L_{\mathbb{H}}(F_5) \otimes_{\mathbb{R}} \mathbb{C} \cong L(F_5) \otimes_{\mathbb{C}} M_2(\mathbb{C}) \cong M_2(L(F_5)) \cong L(F_2),$$

and $L_{\mathbb{R}}(F_2) \otimes_{\mathbb{R}} \mathbb{C} \cong L(F_2)$ by definition. ■

Remark 4.4. Complexification does not reflect isomorphism: $\mathbb{H} \ncong M_2(\mathbb{R})$, yet both complexify to $M_2(\mathbb{C})$. Proposition 4.3 therefore leaves open whether $L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$, which is part of Problem 2. It does not show that they are distinct.

Conjecture 4.5 (β-invariance of free entropy dimension). Let H_N^β be the self-adjoint $N \times N$ matrices over $\mathbb{R}, \mathbb{C}, \mathbb{H}$ for $\beta = 1, 2, 4$, of real dimension

$$d_\beta(N) = N + \beta \frac{N(N-1)}{2}.$$

Define the β-microstate free entropy χ_β with $\log \operatorname{vol} \Gamma_\beta$ normalised by $d_\beta(N)$, and the modified dimension $\delta_{0,\beta}$ as in [9]. Then, for every finite self-adjoint tuple X with Connes-embeddable moments,

$$\delta_{0,1}(X) = \delta_{0,2}(X) = \delta_{0,4}(X).$$

Status. The case $\beta = 2$ is Voiculescu's theory [9]. The conjecture is open; it is entry point 2 of Problem 2.

Why the earlier "Theorem 4.3" is withdrawn.

- *It is false for $\beta = 2$.* It claimed $\delta^* = k$ for every generating tuple, but $\delta_0(S_1, S_2, S_1) = 2 < 3$ for free semicirculars S_1, S_2.
- *The volume comparison fails.* Microstate spaces for different β have different real dimension ($d_1 \neq d_2 \neq d_4$), so the log-volumes differ at order $N^2 \log N$, not $O(1/N)$.
- *The corrections were misattributed.* The quoted $+5/N$ and $-2.5/N$ are corrections to moments, not to volumes.

Question 4.6 (sub-leading β-discrepancy). With finite-N quantities $\chi_\beta^{(N)}(X; m, \varepsilon)$ normalised as in Conjecture 4.5, does

$$\lim_{N \to \infty} N \Big[\chi_\beta^{(N)}(X; m, \varepsilon) - \chi_2^{(N)}(X; m, \varepsilon) \Big], \qquad \beta \in \{1, 4\},$$

exist (with suitable limits in m and ε), and does it distinguish real forms? This replaces the earlier draft's "free Betti number". That definition multiplied an N-independent limit by N, so it was ill-defined, and the name suggested a link to L^2-Betti numbers that does not exist.

5. Truncation, symmetry and continuum limits (Problem 1)

Definition 5.1. For a lattice graph Λ with link set $E(\Lambda)$ and compact gauge group G, the Kogut–Susskind Hilbert space [10] is $\mathcal{H}_{\text{gauge}} = \bigotimes_{\ell \in E(\Lambda)} L^2(G)$. Let $P_{j_{\max}}$ be the orthogonal projection onto representations of spin at most $j_{\max}$ on every link, and $Q = I - P_{j_{\max}}$. A qubit encoding is an isometry $V : P_{j_{\max}} \mathcal{H}_{\text{gauge}} \to (\mathbb{C}^2)^{\otimes n}$, which requires $\dim P_{j_{\max}} \mathcal{H}_{\text{gauge}} \leq 2^n$.

Proposition 5.2 (truncation preserves positivity). Let θ be a positive semidefinite Hermitian form on $\mathcal{H}_{\text{gauge}}$. Then for every A,

$$\langle \theta A, A \rangle = \langle \theta PA, PA \rangle + 2 \operatorname{Re} \langle \theta PA, QA \rangle + \langle \theta QA, QA \rangle,$$

and $\langle \theta PA, PA \rangle \geq 0$. Truncation of states alone therefore cannot violate positivity.

Proof. Expand $A = PA + QA$ sesquilinearly. PA is itself a vector of $\mathcal{H}_{\text{gauge}}$, so positivity applies to it. ■

© 2026 Roger Cooperman; A.E. Petrunina V.V. · MIT License

Any obstruction to Osterwalder–Schrader positivity [11] must come from the truncated dynamics: the transfer matrix $e^{-aH_{j_{\max}}}$ and its relation to a Euclidean lattice measure. It cannot come from the projection of states.

Question 5.3 (Problem 1, entry point 1). Is $\mathbb{Q}$-structure, $[H_{j_{\max}}, J_n] = 0$ for the encoded Hamiltonian, compatible with reflection positivity at finite lattice spacing? The earlier draft's "Theorem 5.2" claimed a general obstruction; it is withdrawn.

Proposition 5.4 (constraints on $\mathbb{Q}$-symmetric encodings). Let H_n be an encoded Hamiltonian with $[H_n, J_n] = 0$.

1. H_n contains only even-weight Pauli terms (Corollary 1.5).
2. If n is odd, every level, including the ground level, has even multiplicity (Corollary 1.6). The gap must then be measured from the ground multiplet: $\Delta_n = E_{\text{next}} - E_0$, where E_{next} is the next distinct eigenvalue. This is the second part of Exercise 10 (Section 1).
3. If n is even, the symmetry imposes no degeneracy: the ground level may be unique or degenerate.

Conjecture 5.5 (parity independence; Problem 1). Along a renormalisation trajectory with $a \to 0$ and $n \to \infty$, the gaps computed on odd-n and even-n encodings converge to the same limit $\Delta > 0$.

This is what Problem 1 asks, and it is open. The earlier draft's "Lemma 5.3" claimed the opposite: that the odd and even limits differ. Its argument assumed a unique vacuum for even n, which is not implied, and it used the gap formula $E_2 - E_1 = g^2 C_f / j_{\max}$, which is not derived and tends to 0 as $j_{\max} \to \infty$. It is withdrawn.

Proposition 5.6 (J-compatible embeddings). An isometry $\iota : (\mathbb{C}^2)^{\otimes n} \to (\mathbb{C}^2)^{\otimes n'}$ with $\iota J_n = J_{n'} \iota$ exists if and only if $n' - n$ is even and non-negative.

Proof. If such an ι exists, then $\iota J_n^2 = J_{n'}^2 \iota$, that is $(-1)^n \iota = (-1)^{n'} \iota$, and $\iota \neq 0$ forces $n' \equiv n \pmod 2$. Conversely, $\varphi = (|01\rangle + |10\rangle)/\sqrt{2}$ satisfies $J_2 \varphi = (\sigma_y \otimes \sigma_y)\overline{\varphi} = \varphi$. The map $\iota(x) = x \otimes \varphi^{\otimes (n'-n)/2}$ is an isometry, and

$$J_{n'} \iota(x) = S_n \overline{x} \otimes \left(S_2 \overline{\varphi}\right)^{\otimes (n'-n)/2} = J_n x \otimes \varphi^{\otimes (n'-n)/2} = \iota(J_n x). \qquad \blacksquare$$

In particular, no non-zero map intertwines J_n and J_{n+1}. A J-compatible inductive chain must therefore add qubits two at a time.

Remark. Along a chain of J-compatible isometries ι_n, a family of forms with $\iota_n^* \theta_{n+2} \iota_n = \theta_n$ takes the same value on x and on $\iota_n(x)$. Positivity on the inductive limit is then equivalent to positivity at every finite stage; it is not obtained by passing to a limit.

Conjecture 5.7 (inductive $\mathbb{Q}$-regularisation; Problem 1). There exist J-compatible isometric chains $(\mathbb{C}^2)^{\otimes n} \hookrightarrow (\mathbb{C}^2)^{\otimes (n+2)} \hookrightarrow \cdots$ and $\mathbb{Q}$-symmetric encoded Hamiltonians H_n whose completed inductive limit carries an SU(2) Yang–Mills theory satisfying the Osterwalder–Schrader axioms, with a mass gap $\Delta > 0$ measured from the ground multiplet.

This is a reformulation of Problem 1, and it is open. The earlier draft's "Theorem 5.4" asserted an "if and only if" characterisation. Its intertwining condition was impossible by Proposition 5.6, and its proof was unfinished. It is withdrawn.

6. Status of every result

Each numbered result falls into one of three tiers: proved in this paper; proved, but resting on cited theorems or stated assumptions; or open. Definitions (1.1, 1.3, 3.1, 4.1, 5.1) are not listed. Statement 3.3 records Problem 3 as posed; Theorem 3.11 answers it.

Proved in this paper

Result	Content
Theorem 1.2	$J_n^2 = (-1)^n I$
Lemma 1.4	Odd-weight criterion for $\mathbb{Q}$-structured rotations (for all θ)
Corollary 1.5	$\mathbb{Q}$-symmetric Hamiltonians contain only even-weight Pauli terms
Corollary 1.6	Kramers degeneracy for odd n (with the solution of Exercise 10)
Lemma 2.1	$\dim \mathfrak{k}_n = (4^n - (-2)^n)/2$ and sp/so type (solution of Exercise 3)
Lemma 2.2	Majorana degrees: $Z_j Z_{j+1} Z_{j+2}$ has degree 6
Proposition 2.3	ZZZ circuits: $\mathbb{Q}$-structured, abelian DLA of dimension $n-2$, trivially sampleable, not quadratic (solution of Exercise 7)
Lemma 3.4	Degree rule for Majorana monomials
Proposition 3.5	Output probabilities need monomials of every even degree
Theorem 3.6	Block factorisation for disjoint qubit supports
Theorem 3.7	**Exponential DLA with exact linear-time sampling: Problem 3, necessity fails (unconditional)**
Lemma 3.8	Parity padding: $\mathbb{Q}$-structure costs one ancilla
Theorem 3.12	Code criterion for efficient sampling
Proposition 5.2	Truncation of states preserves positivity
Proposition 5.4	Constraints on $\mathbb{Q}$-symmetric encodings
Proposition 5.6	J-compatible embeddings exist if and only if $n' - n$ is even

Proved, resting on cited theorems or assumptions

Result	Content	Depends on
Theorem 3.2	Known simulation results (matchgates, g-sim)	Cited [3–6]
Theorem 3.9	**Padded IQP: abelian DLA of dimension $n(n+1)/2$, hard to sample**	The DLA and the output equality are proved here; hardness cites [12] (no collapse of the polynomial hierarchy) and [13] (Conjecture 2 of [13], for ℓ_1 error)
Corollary 3.10	The Lie algebra alone cannot decide sampling complexity	Same as Theorem 3.9
Theorem 3.11	**Answer to Problem 3: no**	Part 1 unconditional; part 2 as Theorem 3.9
Theorem 4.2	Amplification formula	Cited [7, 8]
Proposition 4.3	$M_2(L(F_k)) \cong L(F_{(k+3)/4})$; complexification identity (solution of Exercise 6)	Theorem 4.2

Open, stated as conjectures or questions

Result	Question
Question 3.13	Does polynomial dimension of the module generated by ρ_0 imply efficient sampling?
Remark 4.4	Is $L_{\mathbb{H}}(F_5) \cong L_{\mathbb{R}}(F_2)$?
Conjecture 4.5	β-invariance of free entropy dimension
Question 4.6	Sub-leading β-discrepancy
Question 5.3	Is $\mathbb{Q}$-structure compatible with reflection positivity?
Conjecture 5.5	Parity independence of the continuum gap
Conjecture 5.7	Inductive $\mathbb{Q}$-regularisation of Yang–Mills

Appendix A. Errata relative to the chained draft

Numbering refers to the chained draft *The $\mathbb{Q}$-Structured Simulability Dichotomy: Antiunitary Parity, Real Forms and Continuum Limits* of 29 October 2026, the predecessor of this paper.

Draft result	Verdict	Correction
Thm 1.2: "J_n is an isometric involution"	Wording error	$J_n^2 = -I$ for odd n, so J_n is an involution only for even n; the redundant K^2 step is removed (Theorem 1.2).
Lemma 1.4: rotation in K_n iff $w(P)$ odd	Needs "for all θ"	At $\theta = 2\pi$ an even-weight rotation equals $-I \in K_n$; the qualifier and the derivative argument are added (Lemma 1.4).
Lemma 2.1: dimension formula	Correct	The identification with sp/so is now justified, not only matched by dimension (Lemma 2.1).
Lemma 2.2: degree 6, with $\Phi(X_j) = i\gamma_{2j-1}\gamma_{2j}$	Statement correct; proof mixed conventions	One convention (Z on site) throughout; X_j has degree $2j-1$; with X on site, ZZZ has degree 5 (Lemma 2.2).
Thm 3.2: polynomial DLA "sufficient but not necessary"	Unproved; counterexample invalid	Sufficiency is open; the counterexample assumed an exponential block leaves $\|0\rangle$ fixed, which fails; the displayed relations ($\nmid$, $\not\Rightarrow$ intractability) had the wrong form. The correct answer is Theorem 3.11: necessity fails unconditionally (Theorem 3.7), sufficiency fails unless the polynomial hierarchy collapses (Theorem 3.9).
Lemma 3.3: "adjoint action preserves degree 6"	False	Degree-6 brackets reach degree 10 (Lemma 3.4); $\dim \mathcal{H}_{\text{state}} = 2^n$, not n^{2n}; all X_j were mapped to one monomial. The correct content is Proposition 3.5.

Draft result	Verdict	Correction
Thm 3.5: disconnected commutation graph ⇒ tensor factorisation	False as stated	Commuting generators may share qubits (XX, ZZ); the correct hypothesis is disjoint supports, under which the DLA is polynomial (Theorem 3.6).
Lemma 4.2: "structurally distinct real forms", "absolute classification failure"	Computation correct; claims overstated	Non-isomorphism is open (Proposition 4.3, Remark 4.4).
Thm 4.3: $\delta_1^* = \delta_2^* = \delta_4^* = k$	False	Counterexample (S_1, S_2, S_1); volume comparison across different dimensions; now Conjecture 4.5.
Def 4.4: "free Betti number"	Ill-defined	Replaced by finite-N Question 4.6.
Def 5.1: $L^2(G)^{\otimes\Lambda}$	Imprecise	Tensor product over links $E(\Lambda)$; encoding made explicit (Definition 5.1).
Thm 5.2: truncation violates reflection positivity	False	Contradicts its own hypothesis; cross terms omitted; replaced by Proposition 5.2 and Question 5.3.
Lemma 5.3: odd and even gap limits differ	Unproved; premise false	Even n does not force a unique vacuum; the gap formula is underived; replaced by Proposition 5.4 and Conjecture 5.5.
Thm 5.4: inductive regularisation, "if and only if"	Impossible as stated; proof unfinished	$\iota J_n = J_{n+1}\iota$ forces $\iota = 0$; embeddings must step by two (Proposition 5.6); now Conjecture 5.7.

Appendix B. Numerical verification

The validation scripts (Python 3 and NumPy, MIT License) are in the GitHub repository listed on the front page. There is one script per section of this paper plus one for Appendix A; `python run_all.py` runs all 190 checks (Section 1: 34, Section 2: 40, Section 3: 52, Section 4: 18, Section 5: 25, Appendix A: 21), and all pass. The README maps every numbered result to its checks; cited theorems and open questions are marked as notes rather than checks.

Check	Result
$J_n^2 = (-1)^n I$, $n = 1, \ldots, 7$	Confirmed
$J_n P J_n^{-1} = (-1)^{w(P)} P$ for all Pauli strings, $n = 2, 3$	Confirmed
Even-weight rotation at $\theta = 2\pi$ is J-fixed	Confirmed
Random $\mathbb{Q}$-symmetric H: paired spectrum for $n = 1, 3, 5$; simple spectrum for $n = 4$; $-Z_1 Z_2$ degenerate	Confirmed
$\dim \mathfrak{k}_n$ formula and sp/so dimensions, $n = 1, \ldots, 10$	Confirmed
$Z_j Z_{j+1} Z_{j+2}$ degree 6 (Z on site), degree 5 (X on site); X_2 degree 3	Confirmed
ZZZ DLA $= n - 2$, abelian, $n = 3, \ldots, 8$; output $0 \cdots 0$ with probability 1	Confirmed

Check	Result
Degree rule on 300 random monomial pairs	Confirmed
$\|x\rangle\langle x\|$ has all 2^n components $\pm 2^{-n} Z_S$	Confirmed
Block factorisation on a 40-gate circuit	Confirmed
XX, ZZ output does not factorise	Confirmed: $(0.585, 0, 0, 0.415)$
Positivity expansion and $\langle \theta PA, PA\rangle \geq 0$	Confirmed
$J_n \to J_{n+1}$ intertwiners: only zero; $J_n \to J_{n+2}$: $x \otimes \varphi$ works	Confirmed, $n = 1, 2, 3$
Theorem 3.7: $\dim \mathfrak{g}_n^A = 3 \cdot 2^{n-2}$ by exact Pauli closure	Confirmed, $n = 3, \ldots, 12$ (up to 768)
Theorem 3.7: full statevector equals the single-qubit prediction	Confirmed, 30 random 60-gate circuits, $n = 4, 6, 8$
Lemma 3.8: same distribution, $\dim \mathfrak{g} \leq \dim \mathfrak{g}' \leq 2 \dim \mathfrak{g}$	Confirmed, 40 random generator sets
Theorem 3.9: padded Ising IQP is abelian of dimension $n(n+1)/2$, reproduces the IQP distribution, and $\langle 0\|C\|0\rangle = Z(e^{i\pi/8})/2^n$	Confirmed, $n = 3, 4, 5$
Theorem 3.12: state supported on the code C	Confirmed, 5 random generator sets
Question 3.13: module dimensions in the table of Section 3.2	Confirmed

References

[1] R. Cooperman and A.E. Petrunina V.V., *The Cooperman–Petrunina Challenge Series: Announcement, Primer and Reading Guide*, Zenodo (2026), https://doi.org/10.5281/zenodo.23423138.

[2] F. J. Dyson, "The threefold way," J. Math. Phys. 3, 1199–1215 (1962).

[3] L. G. Valiant, "Quantum circuits that can be simulated classically in polynomial time," SIAM J. Comput. 31, 1229–1254 (2002).

[4] B. M. Terhal and D. P. DiVincenzo, "Classical simulation of noninteracting-fermion quantum circuits," Phys. Rev. A 65, 032325 (2002).

[5] R. Jozsa and A. Miyake, "Matchgates and classical simulation of quantum circuits," Proc. R. Soc. A 464, 3089–3106 (2008).

[6] M. L. Goh, M. Larocca, L. Cincio, M. Cerezo and F. Sauvage, "Lie-algebraic classical simulations for variational quantum computing," arXiv:2308.01432.

[7] K. Dykema, "Interpolated free group factors," Pacific J. Math. 163, 123–135 (1994).

[8] F. Rădulescu, "Random matrices, amalgamated free products and subfactors of the von Neumann algebra of a free group, of noninteger index," Invent. Math. 115, 347–389 (1994).

[9] D. Voiculescu, "The analogues of entropy and of Fisher's information measure in free probability theory II," Invent. Math. 118, 411–440 (1994).

[10] J. Kogut and L. Susskind, "Hamiltonian formulation of Wilson's lattice gauge theories," Phys. Rev. D 11, 395–408 (1975).

[11] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," Comm. Math. Phys. 31, 83–112 (1973).

[12] M. J. Bremner, R. Jozsa and D. J. Shepherd, "Classical simulation of commuting quantum computations implies collapse of the polynomial hierarchy," Proc. R. Soc. A 467, 459–472 (2011), https://arxiv.org/abs/1005.1407.

[13] M. J. Bremner, A. Montanaro and D. J. Shepherd, “Average-case complexity versus approximate simulation of commuting quantum computations,” Phys. Rev. Lett. 117, 080501 (2016), https://arxiv.org/abs/1504.07999.

Licence

MIT License

Copyright (c) 2026 Roger Cooperman; A.E. Petrunina V.V.

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the “Software”), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED “AS IS”, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

In this paper, “Software” covers the text of the paper and the accompanying validation scripts.